3.4 Derivatives of Inverse Trigonometric Functions

You missed this lesson. Copy this into your notes. That is all you need to do.

Question of the day

How fast does an angle change when its sine, or its tangent, changes?

Targets

- I can say what arcsin, arccos and arctan mean.
- I can find the derivative of an inverse trig function, with the chain rule.
- I can find a slope or a tangent line for an inverse trig curve.

Concepts

- **What arcsin means** ($y = \arcsin x$? $\sin y = x$): arcsin x is the angle whose sine is x. $\sin^{-1} x$ means arcsin x, not $1/\sin x$.
- **The three rules** ($\frac{d}{dx} \arcsin x = 1/\sqrt{(1 - x^2)}$ $\frac{d}{dx} \arctan x = 1/(1 + x^2)$): arcsin and arccos have a square root. arctan has no root and a plus sign.
- **The sign pair** ($\frac{d}{dx} \arccos x = -1/\sqrt{(1 - x^2)}$): arccos has the opposite sign of arcsin. Everything else is the same.
- **With the chain rule** ($\frac{d}{dx} \arcsin u = u? \ / \ \sqrt{(1 - u^2)}$): Put u in for x, then multiply by u?. Square the whole inside: $(3x)^2 = 9x^2$.
- **Where they come from** ($1 = \cos y \cdot dy/dx$): Write $x = \sin y$ and differentiate. A right triangle with hypotenuse 1 turns cos y into $\sqrt{(1 - x^2)}$.

What the class predicted, and what is true

- Sine has a flat tangent at its peak. Mirror that piece across $y = x$. At $x = 1$, the arcsin curve has a tangent that is... **vertical.** Rise and run swap in a mirror. A flat tangent becomes a vertical tangent, so the slope of arcsin grows without bound near $x = 1$.
- The graph of arctan levels off for large x. As x gets larger, its slope gets... **closer to 0.** A curve that levels off has a tangent that gets flatter and flatter, so the slope heads toward 0.

Huddle notes (answers filled in)

Definitions

- **arcsin x:** The **angle** whose sine is x. It is not $1/\sin x$.
- **Inverse rule:** If $y = \arcsin x$, then $\sin y =$ **x**. Differentiate both sides to find dy/dx.
- **The three rules:** arcsin: $1/\sqrt{(1 - x^2)}$. arccos: $-1/\sqrt{(1 - x^2)}$. arctan: $1/(1 + \mathbf{x^2})$.
- **Chain rule:** Put u in for x, then multiply by **u?**. Square the **whole inside**.

Worked example

Differentiate y = arcsin(3x).

- The inside is u = **3x**
- Its derivative is u? = **3**
- Square the whole inside: u^2 = **$9x^2$**
- So dy/dx = 3 / $\sqrt{(1 - 9x^2)}$

Label the triangle

1. The longest side has length **1**.
2. The side opposite y has length **x**.
3. The third side has length **$\sqrt{(1 - x^2)}$**.

One worked sample of each skill

Differentiate

Level 1. Differentiate.
Answer: $\frac{-1}{\sqrt{1-x^2}}$ Use the arccos rule directly.

Level 2. Differentiate.
Answer: $\frac{5}{\sqrt{1-(5x)^2}}$ The inside is $u = 5x$ and $u' = 5$. Put them in the arcsin rule.

Level 3. Differentiate.
Answer: $\frac{2x}{1+(x^2)^2}$ The inside is $u = x^2$ and $u' = 2x$. Put them in the arctan rule.

Level 4. Differentiate.
Answer: $\frac{\frac{-1}{x^2}}{1+\left(\frac{1}{x}\right)^2}$ The inside is $u = \dfrac{1}{x}$ and $u' = -\dfrac{1}{x^2}$. Put them in the arctan rule.

Slope at a point

Level 1. $y = \arctan x$. Find $\dfrac{dy}{dx}$ at $x = 3$.
Answer: 0.1 $u(3) = 3$ and $u'(3) = 1$. Put them in the arctan rule: the slope is 0.1.

Level 2. $y = \arcsin(5x)$. Find $\dfrac{dy}{dx}$ at $x = 0.12$.
Answer: 6.25 $u(0.12) = 0.6$ and $u'(0.12) = 5$. Put them in the arcsin rule: the slope is 6.25.

Level 3. $y = \arcsin(0.6e^x)$. Find $\dfrac{dy}{dx}$ at $x = 0$.
Answer: 0.75 $u(0) = 0.6$ and $u'(0) = 0.6$. Put them in the arcsin rule: the slope is 0.75.

Level 4. $y = \arcsin(0.6e^x)$. Find $\dfrac{dy}{dx}$ at $x = 0$.
Answer: 0.75 $u(0) = 0.6$ and $u'(0) = 0.6$. Put them in the arcsin rule: the slope is 0.75.

Spot the right rule

Level 1. Which expression is $\dfrac{dy}{dx}$ for $y = \arccos x$?
(A) 1 / (1 + x^2) (B) –1 / $\sqrt{}$(1 – x) (C) 1 / $\sqrt{}$(1 – x^2) **(D) –1 / $\sqrt{}$(1 – x^2)**
Answer: D The inside is $u = x$, so $u' = 1$ and $u^2 = x^2$. The arccos rule gives –1 / $\sqrt{}$(1 – x^2).

Level 2. Which expression is $\dfrac{dy}{dx}$ for $y = \arcsin(5x)$?
(A) 5 / $\sqrt{}$(1 – 25x^2) (B) 5 / (1 + 25x^2) (C) 5 / $\sqrt{}$(1 – 5x) (D) 1 / $\sqrt{}$(1 – 25x^2)
Answer: A The inside is $u = 5x$, so $u' = 5$ and $u^2 = 25x^2$. The arcsin rule gives 5 / $\sqrt{}$(1 – 25x^2).

Level 3. Which expression is $\dfrac{dy}{dx}$ for $y = \arctan\left(x^2\right)$?
(A) 1 / (1 + x?) (B) 2x / (1 + x^2) (C) 2x / $\sqrt{}$(1 – x?) **(D) 2x / (1 + x?)**
Answer: D The inside is $u = x^2$, so $u' = 2x$ and $u^2 = x^4$. The arctan rule gives 2x / (1 + x?).

Level 4. Which expression is $\dfrac{dy}{dx}$ for $y = \arctan(e^x)$?
(A) e? / (1 + e?) (B) e? / $\sqrt{}$(1 – e^2?) **(C) e? / (1 + e^2?)** (D) 1 / (1 + e^2?)
Answer: C The inside is $u = e^x$, so $u' = e^x$ and $u^2 = e^{2x}$. The arctan rule gives e? / (1 + e^2?).

Tangent line

Level 1. Write the tangent line to $y = \arctan x$ at $x = 0$.
Answer: $1\,(x - 0) + 0$ The point is $(0, 0)$ and the slope is 1. So $y = 1(x - 0) + 0$.

Level 2. Write the tangent line to $y = \arctan(4x)$ at $x = 0.25$.
Answer: $2\,(x - 0.25) + \frac{\pi}{4}$ The point is $(0.25, \frac{\pi}{4})$ and the slope is 2. So $y = 2(x - 0.25) + \frac{\pi}{4}$.

Level 3. Write the tangent line to $y = \arctan\left(\dfrac{1}{x}\right)$ at $x = 1$.
Answer: $(-0.5)\,(x - 1) + \frac{\pi}{4}$ The point is $(1, \frac{\pi}{4})$ and the slope is -0.5. So $y = -0.5(x - 1) + \frac{\pi}{4}$.

Level 4. Write the tangent line to $y = \arctan\left(\dfrac{1}{x}\right)$ at $x = 1$.
Answer: $(-0.5)\,(x - 1) + \frac{\pi}{4}$ The point is $(1, \frac{\pi}{4})$ and the slope is -0.5. So $y = -0.5(x - 1) + \frac{\pi}{4}$.